Explorer Academy

Campaign 01 – Explorer Academy

Investigate the abandoned Outpost Echo and become a fully qualified Explorer.

Theme: Scientific Exploration, Mystery, Engineering, Observation, Critical Thinking

Recommended age: 9-10

Estimated duration: 10-12 Weeks (Adaptive)

This workbook is a printable companion to Explorer Academy. Every mission remains fully completable with nothing but a blank notebook — nothing here is required to progress. Sections marked *optional* are exactly that.

Welcome to Explorer Academy

Explorer Academy has been designed differently from traditional educational programmes. Your child will experience stories, mysteries, investigations and engineering challenges. Behind every activity is carefully planned curriculum coverage, but this is intentionally hidden from the learner. Your role is not to teach. Your role is to support exploration.

Your Role: Mission Control

Parents act as Mission Control, providing encouragement and practical support rather than direct instruction.

- Choosing an appropriate session length
- Providing simple materials
- Helping prepare optional experiments
- Encouraging discussion after missions
- Celebrating persistence and curiosity

Avoid solving problems for your child. When they ask for help, respond with questions that encourage further thinking.

Questions that help without giving answers

- What evidence do you have?
- How could you test that idea?
- What else might explain it?
- What changed your mind?

Session Length & Materials

Session length guide

Available time	What the Explorer experiences
30 minutes	Core activities only
45 minutes	Core + selected Extension
60 minutes	Standard mission
90 minutes	Core + Extension + Rabbit Holes

Materials

Essential

- Notebook or sketchbook
- Pencil
- Ruler
- Eraser

Useful

- Coloured pencils
- Tape
- Scissors
- Glue
- Simple measuring tape
- Torch
- Household timer

Occasionally used

- Cardboard
- String
- Paper cups
- Elastic bands
- Coins
- Kitchen scales
- Measuring jug

Supporting Your Child

Helpful

- Encourage observation
- Ask open questions
- Celebrate good reasoning
- Allow mistakes
- Discuss discoveries

Avoid

- Giving answers immediately
- Correcting every mistake
- Treating activities as tests
- Comparing progress with others

Assessment Philosophy

Explorer Academy does not rely on traditional tests. Progress is demonstrated through the learner's ability to ask increasingly sophisticated questions, make evidence-based decisions, explain reasoning clearly, revise conclusions when appropriate, plan investigations independently, and apply previously learned ideas to unfamiliar situations. Success is measured by growth in investigative thinking.

Curriculum Mapping

Kept out of the Explorer's own experience by design — this section is for parents and guardians only.

Reading

England (United Kingdom) — Comprehension, Inference and Evidence Evaluation

- Retrieval
- Inference
- Sequencing
- Vocabulary in context
- Summarising
- Comparing information from multiple sources
- Reading increasingly complex informational texts
- Emphasised in Missions 1, 4, 8, 12, 14, 19

Writing

England (United Kingdom) — Investigative and Reflective Writing

- Field notes
- Observations
- Instructions
- Reports
- Explanations
- Predictions
- Reflections
- Evidence summaries
- Emphasised throughout all missions, with increasing independence

Mathematics

England (United Kingdom) — Measurement, Data and Problem Solving

- Number
- Measurement
- Time
- Geometry

- Coordinates
- Data handling
- Tables
- Graphs
- Estimation
- Multi-step problem solving
- Emphasised in Missions 3, 6, 7, 10, 11, 12, 13

Science

England (United Kingdom) — Working Scientifically

- Working scientifically
- Fair testing
- Observation
- Classification
- Materials
- Forces
- Electricity
- Earth science
- Weather
- Space
- Environmental systems
- Scientific enquiry is embedded across the entire campaign rather than confined to individual lessons

Engineering & Design

England (United Kingdom) — Design, Test and Evaluate

- Plan
- Build
- Test
- Improve
- Evaluate
- Communicate design decisions
- Major engineering missions: 7, 11, 15, 17

Cross-Curricular Skills

England (United Kingdom) — Critical Thinking and Communication

- Critical thinking
- Communication
- Creativity
- Collaboration (through discussion)
- Problem solving
- Information literacy
- Reflection
- Resilience
- Reinforced repeatedly through authentic investigative contexts rather than standalone activities

Mission-by-Mission Workbook

Mission 1: The Invitation

Notebook Instructions

Set up your Explorer Journal

This is where your Explorer Journal begins — you'll use it in every mission from now on.

1. On the very first page, write **Explorer Journal** at the top, and your Explorer name underneath it. This is your title page — leave the rest of it blank.
2. Turn to the next page. That's where Mission 1 starts.

Read the invitation

After reading Explorer Academy's invitation, write a short list on your journal page of 2–3 things it says Explorer Academy is looking for in a future Explorer. A few words per point is enough — it doesn't need full sentences.

Record your expectations

Underneath that list, write a few sentences answering: *What do you expect to find at Outpost Echo?* Write what you honestly think, not what sounds most impressive.

First Observation Challenge

Choose any ordinary object near you. On a fresh page, number down the margin from 1 to 5, and write one careful observation next to each number. Push past the first, obvious thing you notice — good Explorers look twice.

Extension (optional) — Design a Personal Explorer Emblem

On a fresh page, sketch a small emblem — like a badge or crest — that represents you as an Explorer. Symbols, initials, colours: whatever feels right.

Extension (optional) — Research Famous Explorers

Pick one real explorer from history. Write their name and three interesting facts about their expeditions.

Rabbit Hole (optional) — What Makes an Explorer?

No journal entry needed for this one. Talk with someone else about the question: what makes someone an explorer rather than simply a traveller?

Words worth knowing

- **Probationary** — a trial period, where you have to show you can do something well before it becomes permanent. That's why Outpost Echo is a *probationary* expedition: it's Explorer Academy's way of seeing if you're ready.

Answer Guide — what a strong response looks like

- The journal's first entry names at least 2–3 specific qualities Explorer Academy says it's looking for, not a vague restatement of the letter.
- The recorded expectations are genuine and specific, even if uncertain or ordinary — not a generic "I expect it to be exciting."
- The five observations go beyond the first, obvious detail — at least a couple describe something a first glance would miss.

Mission 2: Arrival at Outpost Echo

Notebook Instructions

Explore the station map

As you walk through Outpost Echo's map with Atlas, list its main areas in your journal — one line each (for example: laboratory, observatory, living quarters).

Label the important areas

Next to each area on your list, add a short note of what it's used for, based on what you've seen so far.

Record unusual observations

On a fresh page, write down anything about the station that seems unusual or out of place — equipment left out, notes left unfinished, anything that makes you pause.

Draw your sketch map

In your journal, draw a simple sketch map of Outpost Echo from your own survey. Use a ruler for straight walls or paths if you have one, but a freehand sketch is fine too — it just needs to be clear enough for you to read later.

Extension (optional) — Improve the station map

Add symbols to your sketch map (a small square for a door, a wavy line for water, and so on), and draw a key in the corner explaining what each symbol means. If you'd like a printed reference for common map symbols, see the optional printable for this mission.

Rabbit Hole (optional) — How Are Research Stations Organised?

No journal entry needed. Find out how a real scientific research station is organised, and compare it with what you've seen of Outpost Echo.

Optional Printable

Optional map-symbols reference card — a small set of cartography symbols is faster to recognise from a printed card than to hand-copy, and it's explicitly offered as an optional extra for the Extension activity, not the core sketch map.

Mission 2 — Map Symbols Reference Card (Optional)

If you'd like a printed version, this reference card shows a few common cartography symbols you can copy onto your own sketch map's key. Copying them by hand into your journal works just as well — printing is only for convenience.

Symbol	Meaning
Small square	Door
Wavy line	Water
Dashed line	Path or corridor
Triangle	Stairs
Circle with dot	Point of interest
Compass rose	North direction

Use these as a starting point — Explorers are free to invent their own symbols too, as long as the key explains them clearly.

Answer Guide — what a strong response looks like

- The sketch map labels at least the main areas explored (laboratory, observatory, living quarters, etc.) — it doesn't need to be to scale, just clearly ordered.
- "Unusual observations" name a specific detail (e.g. "a notebook left open mid-sentence") rather than a general feeling ("it felt weird").

Mission 3: Explorer's Toolkit

Notebook Instructions

Compare measuring tools

Gather a few measuring tools you have at home (a ruler, tape measure, measuring jug, kitchen scale — whatever's around). In your journal, write each tool's name and one sentence on what it's best used for.

Estimate, then measure

Choose 3–4 objects nearby. In your journal, draw a table with three columns headed **Object**, **Estimate**, and **Actual**. Guess each object's size first and write it in the Estimate column — before you measure anything. Then measure each object with a tool from your comparison and fill in the Actual column.

Create an observation table

On a fresh page, design your own simple table for recording measurements and observations — headings of your choice, ruled columns and rows. This is the table you'll reuse whenever a future mission asks you to record data.

Practise recording evidence

Pick one thing you've measured or noticed today, and write it down as clearly as you can — precisely enough that someone else reading your journal would understand exactly what you found, without needing to ask you.

Extension (optional) — Invent an additional Explorer tool

Sketch or describe a tool an Explorer might need that doesn't already exist in your toolkit. Explain what problem it solves.

Rabbit Hole (optional) — Real Field Journals

No journal entry needed. Find out how real scientists record their observations in field journals, and compare their approach with yours.

Words worth knowing

- **Estimate** — a sensible guess based on what you already know, made *before* you check with a tool. A good estimate isn't a random number; it's your best thinking without measuring yet.

Answer Guide — what a strong response looks like

- Estimates were written down before measuring — if every estimate matches the actual measurement suspiciously closely, it may have been adjusted afterwards.
- The observation table has clear column headings and consistent units — someone else could read it without asking a follow-up question.

Mission 4: The Silent Logs

Notebook Instructions

Read the recovered logs

As you read each recovered log fragment, jot a one-line note of what it reveals — enough to jog your memory later, not a full copy.

Arrange events chronologically

On a fresh page, draw a vertical timeline down the middle of the page. Place each event from the logs onto the timeline in the order you think it happened, earliest at the top.

Highlight the important evidence

Go back through your notes and mark (underline, star, or circle) the pieces of evidence that matter most to understanding what happened at Outpost Echo.

Separate facts from assumptions

Draw two columns headed **Facts** and **Assumptions**. Sort what you've found into the two columns — a fact is something the logs directly state; an assumption is something you're filling in yourself, even if it feels likely.

Extension (optional) — Rewrite a damaged log entry

Choose one damaged log entry and rewrite it in your journal from a different expedition member's point of view — what might they have seen or felt that the original entry doesn't mention?

Rabbit Hole (optional) — Reconstructing Incomplete Records

No journal entry needed. Find out how historians piece together history from damaged or incomplete records, and compare their methods with your own timeline work.

Words worth knowing

- **Chronological** — arranged in the order things actually happened, earliest first. Your timeline is a chronological record of the expedition's last days.
- **Assumption** — something you believe is probably true, even though nothing directly confirms it. It's different from a fact, which the logs state outright.

Answer Guide — what a strong response looks like

- The chronological order is justified by specific clues from the logs, not just a plausible-sounding guess.
- Each item in the "assumptions" list is something inferred, not something the logs directly state — anything in the "facts" list should be traceable to a specific line in the logs.

Mission 5: Strange Footprints

Notebook Instructions

Measure the tracks

In your journal, draw a table with columns for **Length**, **Width**, and **Spacing**. Measure the footprints as instructed and record your measurements in the table.

Compare the evidence

List each possible explanation Atlas gives you, and underneath each one, note how well your measurements match it — where it fits, and where it doesn't.

Identify similarities and differences

Draw a simple two-column list — **Similarities** and **Differences** — for each possible explanation, comparing it against the evidence you've gathered.

Justify your conclusion

Write a short paragraph explaining which explanation your evidence best supports, and why you're ruling the others out. A good answer names the specific evidence that eliminated each rejected explanation — not just "it didn't feel right."

Extension (optional) — Create a field guide of track patterns

On a fresh page (or a few), sketch a small field guide showing a few different track patterns you can imagine or have seen, with a short note under each about what kind of animal might leave it.

Rabbit Hole (optional) — How Do Wildlife Researchers Identify Animals?

No journal entry needed. Find out how wildlife researchers identify animals from tracks alone, without ever seeing the animal itself.

Words worth knowing

- **Eliminate** — to rule something out because the evidence doesn't support it, not because it merely seems less likely. Explorers eliminate explanations one at a time, using evidence, rather than picking the answer that feels best.

Answer Guide — what a strong response looks like

- Each rejected explanation has a specific reason tied to a measurement — "it didn't fit" isn't enough; "the spacing didn't match" is.
- The surviving conclusion is the one explanation that fits every measurement, not just the most exciting one.

Note: The campaign deliberately doesn't specify a single fixed cause for the footprints — the point of this mission is the elimination process, not landing on one universally "correct" answer. Judge the reasoning, not which explanation was chosen.

Mission 6: Weather Watch

Notebook Instructions

Record today's observations

In your journal, write today's date and note the sky, wind, and anything else you notice about the weather right now.

Measure (or estimate) the temperature

If a thermometer is available, record the temperature. If not, estimate it carefully and write "estimated" next to your reading so you remember it wasn't measured directly.

Create a simple chart

Draw a small chart in your journal to hold your weather data — a table with a row for today works fine, and you can add more rows on future days.

Compare with the archived records

Next to your chart, write a few notes comparing today's weather with the expedition's archived records: what's similar, what's different?

Extension (optional) — Predict tomorrow's weather

Write a short prediction for tomorrow's weather, and underneath it, explain your reasoning — which piece of evidence led you to that prediction.

Rabbit Hole (optional) — How Meteorologists Make Forecasts

No journal entry needed. Find out how meteorologists actually make weather forecasts, and compare their approach with yours.

Words worth knowing

- **Meteorologist** — a scientist who studies weather and makes forecasts, using patterns in evidence rather than guessing.

Answer Guide — what a strong response looks like

- Today's temperature is labelled as measured or estimated — it shouldn't be presented as more precise than it actually is.
- The chart lets you compare today's reading against the archived data at a glance, without needing the surrounding notes to make sense of it.

Mission 7: The Broken Bridge

Notebook Instructions

Analyse the problem and plan your design

In your journal, write down what your bridge needs to support and what materials you have available. Then sketch your bridge design before you build anything — label the parts and how they fit together.

Build, test and refine

This mission's build-and-test step is a full experiment — follow `generated/resources/experiments.json` (Mission 7 entry) for materials, steps and safety notes. Bring your journal to the test: you'll need it for the next step.

Record your improvements

After testing, write down what you changed between your first and final design, and why each change helped. Be specific — "I folded the card into a ridge shape, which stopped it bending as easily" is more useful than "I made it stronger."

Extension (optional) — Compare bridge structures

Look at pictures or examples of a few different real bridge structures (beam, arch, truss). In your journal, write a short explanation of why some shapes hold more weight than others.

Rabbit Hole (optional) — How Engineers Choose Bridge Designs

No journal entry needed. Find out how engineers choose different bridge designs for different environments.

Answer Guide — what a strong response looks like

- Two distinct attempts are recorded, not just one "final" design with no visible history of what came before.
- The explanation of what changed names a specific structural reason (e.g. "folding the card added rigidity"), not just "it worked better."

Expected outcome: A flat, unfolded sheet of card bends and fails quickly under only a little weight. Folded into a ridge, tube or arch shape, the same material holds noticeably more weight before buckling — folding increases rigidity by spreading the load along the fold instead of concentrating it at one weak point (see generated/resources/experiments.json, Mission 7).

Mission 8: Message in the Static

Notebook Instructions

Decode the damaged message

As you go through the corrupted transmission, write down every word or phrase you can make out, in the order it appears.

Complete the missing sections

Underneath your notes, suggest what the missing sections might say, using repeated patterns and context clues from what you *can* read. Write these guesses clearly separate from what's actually legible — for example, put your suggestions in brackets.

Compare possible interpretations

If more than one interpretation of the message seems plausible, write each one down and note which pieces of evidence support it best.

Identify confirmed facts

Draw a short list titled **Confirmed Facts** — only include what the message actually states clearly, not your reconstructed guesses.

Extension (optional) — Create a coded message

Invent a simple coded message of your own, for another Explorer to solve. Write both the coded version and an answer key on a separate page (or fold it under) so it stays a surprise.

Rabbit Hole (optional) — Recovering Damaged Historical Records

No journal entry needed. Find out how scientists and historians recover damaged documents or radio transmissions.

Words worth knowing

- **Transmission** — a message sent using radio or another signal, especially across a distance. Dr. Quinn's transmission travelled through the station's communications array before it was recovered.

Answer Guide — what a strong response looks like

- Guesses for the missing sections point back to a specific pattern or context clue in the parts that are legible.
- The "confirmed facts" list only includes what the message actually states — reconstructed guesses, however plausible, stay out of it.

Mission 9: Mystery Samples

Notebook Instructions

Observe each sample's properties

For each sample, write down what you actually observe — colour, texture, size, and any other detail — rather than what you guess it might be.

Group similar specimens

Draw a short list of the groups you've decided on, and underneath each group heading, list which samples belong there.

Record your evidence

For each group, write the specific observed properties that led you to put those samples together.

Justify your classifications

Underneath your evidence, write a sentence for each group explaining *why* those properties justify grouping the samples together — using more than one property where you can, not just colour alone.

Extension (optional) — Design a classification key

Sketch a simple branching key (a series of yes/no questions) that could sort a new, unfamiliar sample into one of your groups.

Rabbit Hole (optional) — How Biologists Classify New Species

No journal entry needed. Find out how biologists classify newly discovered species.

Words worth knowing

- **Classify** — to sort things into groups based on shared, observable properties and evidence, not just first impressions or appearance alone.

Answer Guide — what a strong response looks like

- Groupings use more than one observed property, not colour alone — check whether two different-coloured samples with the same texture ended up in different groups without explanation.
- Each group's justification names the specific shared properties, not just "these looked similar."

Mission 10: Water Under Pressure

Notebook Instructions

Trace the water flow

Follow the station's pipework diagram and copy the water's path into your journal, step by step, from where it enters the system to where it should end up. If you'd like a printed copy of the diagram to trace directly, see the optional printable for this mission.

Run the experiment

This mission's investigation into where the flow is blocked is a full experiment — follow `generated/resources/experiments.json` (Mission 10 entry) for materials, steps and safety notes.

Interpret the diagrams

Using what your experiment showed, write a note on the diagram (or next to it in your journal) marking where you think the real malfunction is most likely occurring, and why.

Recommend a repair

Write a short, clear recommendation explaining what repair should be made and why — clear enough that Atlas could act on it without asking you to explain further.

Extension (optional) — Design a more efficient water system

Sketch a design for an improved water recycling system for the station. What would you change about how water moves or is reused?

Rabbit Hole (optional) — Water Recycling in Space

No journal entry needed. Find out how water is recycled aboard spacecraft or remote research stations.

Words worth knowing

- **Malfunction** — when part of a machine or system stops working correctly. It's a more precise word than "broken" — a malfunction can be partial, temporary, or hard to see.

Optional Printable

Optional printed copy of the station's water system diagram (IMAGE-0012) — the Core 'Trace Water Flow' and 'Interpret Diagrams' activities explicitly require tracing an existing pipework diagram, meeting the PHILOSOPHY bar for a diagram that's genuinely useful printed. Hand-copying it works just as well; the printable is offered only for convenience.

Mission 10 — Station Water System Diagram (Optional)

If you'd like a printed copy to trace onto or write notes directly on, this describes the diagram referenced in `generated/image-specifications/images.json` (IMAGE-0012). Hand-copying the diagram from the app into your journal works just as well — this is only for convenience.

The diagram shows the station's water recycling loop as a simple flow path:

```
Intake → Filter → Storage Tank → Pump → Distribution Pipes → (used water) → Recycling Unit → back to Filter
```

Trace or copy this path into your journal, then mark on it where you think the blockage or malfunction is most likely occurring, based on your experiment's results.

Answer Guide — what a strong response looks like

- The recommendation names a specific location in the system, using the diagram's own labels, not a general "something's blocked somewhere."
- The reasoning explicitly connects back to what the household experiment actually showed.

Expected outcome: A blockage narrows the space available for water to flow, increasing resistance — water visibly slows, pools, or stops just above the blockage point, and moving the blockage to a different point changes where that happens (see generated/resources/experiments.json, Mission 10).

Mission 11: The Energy Problem

Notebook Instructions

Analyse energy requirements

Draw a table listing each station system Atlas mentions, with a column for how much energy it needs. Fill it in as you work through the mission.

Solve the circuit challenges

Work through the circuit challenges as instructed. In your journal, jot a short note each time you solve one — what did it teach you about how the systems connect?

Prioritise essential systems

Underneath your energy table, rewrite the systems in the order you'd restore them, from most to least essential, given the energy actually available.

Record your reasoning

Write a short paragraph explaining why you ordered the systems that way. A strong answer names the trade-off you made, not just the final order.

Extension (optional) — Design an emergency power plan

Sketch or write a plan for how the station should handle a future power shortage, using what you learned about prioritising systems.

Rabbit Hole (optional) — Electricity in Extreme Environments

No journal entry needed. Find out how remote research stations generate electricity in extreme environments.

Words worth knowing

- **Circuit** — a complete loop that electricity can flow around. If the loop is broken anywhere, nothing at the end of it will work.

Answer Guide — what a strong response looks like

- Each system in the sequence has a stated reason for its position — "because it needs the least power" or "because other systems depend on it," not just a number.
- The sequence respects the actual energy budget calculated earlier in the mission, not a guessed one.

Mission 12: Star Maps

Notebook Instructions

Match constellations and read the star maps

As you match the recorded star patterns to known constellations, list each match in your journal. Then note the positions the expedition recorded from the station's star maps.

Measure the angles

Using a protractor (or the printable star chart for this mission if you'd like one), measure the angles between the recorded star positions and record them in a table: **Pair of Stars** and **Angle**.

Interpret the observations

Write a few sentences explaining what pattern the matched constellations and measured angles suggest the expedition was tracking.

Extension (optional) — Create a seasonal star guide

Sketch a simple guide showing a few constellations visible at different times of year, with a short note on when best to see each one.

Rabbit Hole (optional) — Navigating by the Night Sky

No journal entry needed. Find out how explorers navigated using the night sky before modern technology.

Words worth knowing

- **Constellation** — a group of stars that forms a recognisable pattern in the sky, named after what it seems to resemble.

Optional Printable

Optional printed star chart (IMAGE-0014) for the Core 'Measure Angles' activity — measuring accurately with a protractor is easier on a printed, precisely plotted chart than a hand-copied one. Hand-copying the recorded positions still works; the printable is offered only for convenience.

Mission 12 — Star Chart for Angle Measuring (Optional)

If you'd like a printed copy to measure angles on directly with a protractor, this describes the star chart referenced in `generated/image-specifications/images.json` (IMAGE-0013). Copying the recorded star positions into your journal by hand and measuring between them works just as well — this is only for convenience.

The chart plots the expedition's recorded star positions as simple labelled dots on a grid, close enough to their true relative positions that the angles between them can be measured accurately with a protractor and ruler. Use it to measure the angles the mission asks for, then note your results in your journal as usual.

Answer Guide — what a strong response looks like

- Measured angles are specific numbers, not rounded to convenient values that weren't actually measured.
- The interpretation of what the expedition was tracking is supported by at least two matched positions, not a single coincidental pattern.

Mission 13: Hidden Patterns

Notebook Instructions

Compare the datasets

As you look through the datasets Atlas provides, write a short note on what each one shows on its own, before combining anything.

Create graphs

Draw a simple graph in your journal (a bar chart or line graph both work) that combines the datasets so any pattern becomes easier to see.

Identify the trend

Underneath your graph, write a sentence describing the trend you can now see, now that the data is combined.

Explain your conclusion using evidence

Write a short paragraph explaining what the trend means, pointing to specific evidence from the datasets — not just describing the shape of the graph.

Extension (optional) — Predict future observations

Using the trend you identified, write a prediction for what future observations might show, and explain your reasoning.

Rabbit Hole (optional) — Discovering Hidden Patterns in Data

No journal entry needed. Find out how scientists discover hidden patterns within large datasets.

Words worth knowing

- **Trend** — a general direction something is changing in, only visible when you look at more than one data point together rather than any single measurement on its own.

Answer Guide — what a strong response looks like

- The trend is described specifically (e.g. "readings rise every third week") rather than vaguely ("there's a pattern").

- At least two different datasets are cited as supporting the same trend — a pattern seen in only one dataset isn't yet a trend.

Mission 14: The Missing Notebook

Notebook Instructions

Reassemble the notebook pages

As you read the surviving pages, note in your journal what order you think they go in, and why — what clue told you one page comes before another?

Arrange events chronologically

Draw a timeline of Dr. Quinn's final observations, using the journal entries, maps, timestamps and previous discoveries you've gathered across earlier missions.

Write the missing sections

For each gap in the notebook, write a short passage reconstructing what might belong there, using evidence from the surviving pages and what you already know.

Distinguish facts from inference

Draw two columns headed **Confirmed Facts** and **Reasonable Inference**. Sort your reconstruction between them — a fact is directly stated somewhere; an inference is a conclusion you've drawn by combining evidence, even if you're fairly confident about it.

Extension (optional) — Write another perspective

Write an additional journal entry from the point of view of a different expedition member, describing the same events Dr. Quinn recorded.

Rabbit Hole (optional) — Reconstructing Incomplete Manuscripts

No journal entry needed. Find out how archaeologists and historians reconstruct incomplete manuscripts.

Words worth knowing

- **Inference —** a reasonable conclusion you reach by combining pieces of evidence together. It's stronger than a guess, but it's still not the same as a fact that's directly stated.

Answer Guide — what a strong response looks like

- Reconstructed sections reference specific surviving pages or earlier discoveries, not just "what seems likely."
- The learner can explain the difference between something they inferred and something they're simply hoping is true.

Mission 15: The Signal Tower

Notebook Instructions

Inspect the tower and identify faults

As you inspect the tower's systems, list everything that looks damaged or disconnected. Next to each one, note what you think the specific fault actually is — not just that something's wrong.

Develop a repair sequence

Decide the order you'd fix the faults in, and write a sentence justifying each choice — what makes one fault more urgent to fix than another?

Test and refine your solutions

This mission's testing step is a full experiment — follow `generated/resources/experiments.json` (Mission 15 entry) for materials, steps and safety notes. Test each proposed fix one at a time, refining any that don't fully work before moving to the next fault.

Record your design decisions

Write up the decisions you made throughout the repair and the reasoning behind each one — enough detail that someone reading it later would understand why you did things in that order.

Extension (optional) — Compare alternative repair strategies

Think of a different order you could have fixed the faults in, or a different way to solve one of them. Write a short comparison, and justify the strategy you actually chose.

Rabbit Hole (optional) — Communication Towers in Remote Environments

No journal entry needed. Find out how communication towers operate in remote environments.

Words worth knowing

- **Diagnose** — to work out the specific cause of a problem by testing and gathering evidence, rather than guessing at a fix.

Answer Guide — what a strong response looks like

- Each fault has its own diagnosis and its own test-and-refine cycle — not one overall attempt covering everything at once.
- The repair order is justified by which fault is most critical, not which was easiest to fix first.

Expected outcome: Light (standing in for a signal) needs a clear, aligned path to reach its target — obstruction blocks it almost completely regardless of alignment or distance, misalignment can redirect it away entirely if severe enough, and distance alone dims and spreads it without fully blocking it (see generated/resources/experiments.json, Mission 15).

Mission 16: Secrets Underground

Notebook Instructions

Examine the rock samples

For each rock sample, write down its properties — colour, texture, weight, anything else you notice — the same way you recorded sample properties back in Mission 9.

Compare the geological layers

Draw a simple side-on sketch of the layers you can see in the survey area, and note anything unusual about how they're arranged.

Interpret the survey maps

Write a few sentences describing what the expedition's survey maps tell you about the shape of the surrounding landscape.

Explain your findings using evidence

Write a paragraph explaining what your findings reveal — and make sure it draws on your samples, your layers, and the maps together, not just one of the three.

Extension (optional) — Produce a geological field sketch

Sketch and label the survey area, marking the key features you've identified. See the optional exemplar image for this mission if you'd like an idea of what a clearly labelled field sketch looks like.

Rabbit Hole (optional) — Studying Landscapes Without Disturbing Them

No journal entry needed. Find out how geologists study landscapes carefully, without damaging them.

Words worth knowing

- **Strata** — the layers of rock or soil stacked on top of each other, usually built up over a very long time. Comparing strata can reveal a landscape's history.

Answer Guide — what a strong response looks like

- The explanation draws on at least two of the three evidence sources (samples, layers, maps), not just whichever felt easiest to describe.

- Specific properties or features are named, not general statements like "the rocks looked old."

Mission 17: The Final Experiment

Notebook Instructions

Define your investigation and form a prediction

This mission has no fixed topic — you choose it. In your journal, write the exact question your investigation will try to answer, then write your prediction and explain your reasoning for it.

Plan and run a fair test

This mission's fair test is a full experiment you design yourself — follow `generated/resources/experiments.json` (Mission 17 entry) for the general framework, safety approach and what "fair" means here, since the materials depend entirely on what you're investigating.

Collect your evidence

Carry out your test carefully, recording your evidence in your journal as you go — don't wait until the end to write it all up from memory.

Analyse your results

Write whether your evidence supports your original prediction, and explain why or why not.

Reflect on the limitations

Write a short, honest reflection on what could have made your investigation stronger — every investigation has limitations, including this one.

Extension (optional) — Repeat with a changed variable

Run your investigation again, changing just one thing from your original test. Compare the two sets of results in your journal.

Rabbit Hole (optional) — Is Experimental Evidence Reliable?

No journal entry needed. Find out how scientists decide whether experimental evidence is reliable.

Words worth knowing

- **Fair test** — a test where you change only one thing at a time and keep everything else the same, so you can tell what actually caused any difference in the result.

Answer Guide — what a strong response looks like

- Only one variable changed between the original test and any repeat — if several things changed at once, it isn't a fair test.
- The reflection names a real, specific limitation (not "everything went perfectly") — a specific limitation is a sign of genuine reflection, not a lack of success.

Note: There's no fixed outcome to check here — since the learner chose the investigation topic themselves, judge the reasoning and fairness of the test, not any specific result.

Mission 18: Connecting the Evidence

Notebook Instructions

Review the evidence

Look back through your Explorer Journal from earlier missions. Write a short summary list of the key pieces of evidence you've gathered so far.

Organise your findings

Group your evidence into related clusters — draw a heading for each group and list which pieces of evidence belong there.

Identify relationships

Write a few sentences describing how your groups connect to each other — what links them together?

Support your conclusion with evidence

Write your overall conclusion about what happened at Outpost Echo, and under it, list the specific pieces of evidence that support it.

Evaluate alternative explanations

Write down one or two other explanations you considered, and explain why the evidence supports your conclusion better than those alternatives.

Extension (optional) — Produce an investigation board

Draw your own investigation board: write or sketch your key pieces of evidence around the page and connect related ones with lines, like a detective's wall. See the optional exemplar image for this mission if you'd like an idea of what one can look like.

Rabbit Hole (optional) — How Evidence-Based Explanations Are Built

No journal entry needed. Find out how detectives, archaeologists and scientists build evidence-based explanations.

Answer Guide — what a strong response looks like

- The conclusion addresses evidence from several different earlier missions, not just the most dramatic one or two.
- At least one alternative explanation is named and specifically ruled out, not just dismissed.

Mission 19: Recover the Archive

Notebook Instructions

Read the final report

As you read Dr. Quinn's final report, note down what it reveals in your journal, a point at a time.

Compare original and current evidence

Write a short comparison between what the expedition originally recorded and what you've discovered yourself throughout the campaign — where do they match, and where do they add something new?

Complete the missing archive sections

Using your own discoveries, write the missing sections of the archive as clearly as you can.

Summarise your findings

Write a summary explaining the expedition mystery and what it reveals about Dr. Quinn's intentions — in your own words, as if explaining it to someone who hasn't followed the whole campaign.

Extension (optional) — Write a letter to future Explorers

Write a letter to a future Explorer, explaining why careful investigation mattered so much to this expedition — and to you.

Rabbit Hole (optional) — Discoveries That Took Years to Verify

No journal entry needed. Research a historical scientific discovery that required years of verification before it was accepted.

Words worth knowing

- **Archive** — a protected collection of records kept safe for the future, especially ones considered too important to risk losing.

Answer Guide — what a strong response looks like

- The summary explains why the expedition suspended operations, not just what they did.
- The completed archive sections draw on discoveries the learner actually made earlier in the campaign, not invented details.

Mission 20: Explorer Assessment

Notebook Instructions

Investigate the unfamiliar problem

This mission has no fixed topic or instructions — that's the point. Write down the problem the Academy has presented you with, in your own words, before deciding anything about how to approach it.

Plan your approach

Write a short plan explaining how you'll investigate the problem and which methods you'll use — decide this before you start collecting evidence.

Collect your evidence

Follow your plan and record your evidence in your journal as you go.

Justify your decisions

Write down the reasoning behind the choices you made during the investigation — not just what you did, but why.

Present your conclusions

Write up your conclusions clearly enough that someone else could follow your reasoning from start to finish.

Extension (optional) — Identify improvements

Write a short list of things that could have made your investigation stronger, if you had more time or different resources.

Rabbit Hole (optional) — How Different Experts Investigate

No journal entry needed. Consider how different kinds of experts might approach the same problem in very different ways.

Answer Guide — what a strong response looks like

- The plan was written before evidence was collected, and the learner can explain why they chose that approach.

- Justifications are specific to decisions actually made during the investigation, not generic statements about "being careful."

Note: This mission has no fixed correct answer by design — assess the reasoning and process, matching the mission's own reward (a rank, not a knowledge badge).

Mission 21: Graduation Day

Notebook Instructions

Reflect on your learning journey

Write a reflection on your whole journey through the campaign, from the invitation in Mission 1 to right now. What's changed about how you investigate?

Review your Explorer Journal

Look back through your journal from the very first entry. Note what stands out to you now, especially anything that reads differently than you remembered.

Identify your favourite investigations

List a few of your favourite investigations from the campaign, and write a sentence explaining why each one stood out.

Complete your final expedition report

Write a short report summarising the whole Outpost Echo investigation — what happened, what you found, and how it was resolved.

Receive your Explorer status

If you'd like a printed certificate to personalise, see the optional printable for this mission. Otherwise, design your own certificate directly in your journal — include your Explorer name and today's date.

Extension (optional) — Design your own future expedition

Write or sketch a design for a future expedition of your own: where would you go, and what would you investigate?

Rabbit Hole (optional) — A Question With No Complete Answer

No journal entry needed. Research a real-world scientific question that still has no complete answer.

Optional Printable

An optional Explorer Certificate template — a certificate is a ceremonial keepsake genuinely suited to printing, unlike the campaign's working content, and it's explicitly offered as an alternative to designing *one directly in the journal, not the only path.*

Mission 21 — Explorer Certificate (Optional)

If you'd like a printed certificate to personalise instead of designing your own in your journal, this is a simple template you can fill in by hand.

EXPLORER ACADEMY

Certificate of Achievement

This certifies that

__________ *(Explorer name)*

has successfully completed probationary training at Outpost Echo and is hereby recognised as a

CERTIFIED EXPLORER

Awarded on: ________

Signed: *Director Orion, Explorer Academy*

Designing your own certificate directly in your journal works just as well — this printable is only offered for convenience, and either way, the achievement is the same.

Answer Guide — what a strong response looks like

- The reflection compares who the learner is now against Mission 1 specifically — not just a general "I learned a lot."
- Favourite investigations are named with a specific reason each, not a generic "they were all fun."

A Closing Note

This workbook is optional from start to finish. A blank notebook, a pencil and a ruler are all any mission actually requires. Everything here exists to make that easier — never to replace it.